

The solutions for this sheet must be submitted on Moodle by 21 May, 15:59.

THE FOLLOWING EXERCISE IS WORTH 2 BONUS POINTS.

Exercise T6.1 – Edges in Min Cuts

Consider a network $N = (V, A, c, s, t)$ with $c(e) > 0$ for all $e \in A$. By $\text{flow}(N)$, we denote the value of a maximum flow in N . For an edge $e \in A$, determine which of the following statements are equivalent. In which cases does at least one implication hold?

- (i) Every decrease of $c(e)$ decreases $\text{flow}(N)$.
- (ii) Every increase of $c(e)$ increases $\text{flow}(N)$.
- (iii) There is a minimum s - t -cut in N that contains e . (i.e., there is an s - t -cut (S, T) such that $e \in S \times T$)
- (iv) Every minimum s - t -cut contains e .
- (v) Every maximum flow f in N satisfies $f(e) = c(e)$.

Hint: You may assume that all capacities are integral. In this case, it suffices to consider increases/decreases of 1 in (i) and (ii). (The answers remain the same with or without the integrality assumption.)

THE FOLLOWING EXERCISE DOES NOT GIVE BONUS POINTS.

Exercise T6.2 – 2/3-Cover in Bipartite Graphs

Let $G = (A \cup B, E)$ be a bipartite graph. We call an edge set $U \subseteq E$ a $2/3$ -cover of G if the graph $G' = (A \cup B, U)$ has the following properties: $\deg_{G'}(a) = 2$ for all $a \in A$ and $\deg_{G'}(b) = 3$ for all $b \in B$. (A perfect matching would be a $1/1$ -cover).

Describe a (polynomial time) algorithm that takes as input a bipartite graph $G = (A \cup B, E)$ and outputs either a $2/3$ -cover, or correctly reports that none exists. The partition $A \cup B$ is part of the input.

Hint: Remember that a description of an algorithm should always include a proof of correctness and runtime.